

Xsuite

Lie Groups
Normal Forms
TPSAs

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Chapter 1

Lie Algebra, Lie Groups, and Compositional Maps

Lie methods provide a useful change of viewpoint. Instead of describing a beam-line element only by the map that sends its entrance coordinates to its exit coordinates, one describes an infinitesimal transformation—a **generator**—and obtains the finite map by exponentiation. Linear transfer matrices are the simplest example of this idea. The same language also applies to nonlinear Hamiltonian maps, with the matrix commutator replaced by the Poisson bracket.

1.1 Lie Algebra

1.1.1 Vector spaces and Lie algebras

A vector space V over a field $\mathbb{F}$ (here typically $\mathbb{R}$ or $\mathbb{C}$) is a set whose elements can be added and multiplied by scalars. The result must remain in V , and these operations obey the familiar linear rules: addition is associative and commutative, there is a zero and an additive inverse, and scalar multiplication is associative and distributes over both vector addition and scalar addition. Consequently, if $X, Y \in V$ and $a, b \in \mathbb{F}$, then $aX + bY \in V$. This last closure test is often the most useful quick check in practice.

A **Lie algebra** $\mathfrak{g}$ over $\mathbb{F}$ is a vector space equipped with a binary operation

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \longrightarrow \mathfrak{g}, \quad (X, Y) \longmapsto [X, Y], \quad (1.1)$$

called the **Lie bracket**. For $X, Y, Z \in \mathfrak{g}$ and $a, b \in \mathbb{F}$ it satisfies

$$\begin{aligned} [aX + bY, Z] &= a[X, Z] + b[Y, Z], \\ [Z, aX + bY] &= a[Z, X] + b[Z, Y] \end{aligned} \quad \text{(bilinearity),} \quad (1.2)$$

$$[X, X] = 0 \quad \text{(alternation),} \quad (1.3)$$

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0 \quad \text{(Jacobi identity).} \quad (1.4)$$

Over the fields used here, $\mathbb{R}$ and $\mathbb{C}$, alternation is equivalent to antisymmetry, $[X, Y] = -[Y, X]$. Writing the axiom as alternation is also correct in characteristic 2, where a minus sign does not distinguish antisymmetry from symmetry. The bracket therefore

measures non-commutativity. If every bracket vanishes, the algebra is called abelian. Notice that a Lie algebra has two quite different operations: vector addition and the bracket. The bracket is not in general an associative multiplication.

1.1.2 Lie groups and representations

A **Lie group** G is both a group and a smooth manifold, with smooth group multiplication and inversion. Its Lie algebra is the tangent space at the identity,

$$\mathfrak{g} = T_e G. \quad (1.5)$$

A tangent vector $X \in T_e G$ is the initial velocity of a smooth curve $g(t)$ through the identity: $g(0) = e$ and $\dot{g}(0) = X$. Here t is simply a real curve parameter, defined at least on an open interval around $t = 0$; it need not represent physical time. Different curves with the same initial velocity represent the same tangent vector. Translating this vector from the identity to every other group element produces a left-invariant vector field. The usual commutator of these vector fields, evaluated back at e , defines the Lie bracket. Thus the algebra contains the infinitesimal transformations of the group. It determines the group locally near the identity, but does not by itself contain all global topological information.

The group itself is usually **not** a vector space. Rotation matrices, for example, form a group under matrix multiplication, but the sum of two rotation matrices need not be a rotation matrix, and multiplying a rotation matrix by zero leaves the group. This is precisely why it is useful to move to the tangent space: $T_e G$ is a vector space even when G is curved.

If $\mathfrak{g}$ is finite-dimensional, choose a basis $t_1, \dots, t_n$. These basis elements are commonly called the **generators**. Every infinitesimal transformation is

$$X \in \mathfrak{g} \Leftrightarrow X = x^a t_a, \quad (1.6)$$

where repeated indices are summed, and closure of the bracket gives

$$[t_a, t_b] = f_{ab}^c t_c. \quad (1.7)$$

The numbers f_{ab}^c are the structure constants in this basis. Antisymmetry and the Jacobi identity impose corresponding relations on them. A different basis changes the structure constants but not the Lie algebra. The dimension of $\mathfrak{g}$ equals the dimension of G as a manifold.

A group or algebra can act on many different spaces. A **representation** of a group on a vector space V is a homomorphism¹

$$\rho : G \longrightarrow \mathrm{GL}(V), \quad \rho(g_1 g_2) = \rho(g_1) \rho(g_2), \quad (1.8)$$

and a representation of its algebra is a linear map²

$$\rho_* : \mathfrak{g} \longrightarrow \mathfrak{gl}(V), \quad \rho_*([X, Y]) = \rho_*(X) \rho_*(Y) - \rho_*(Y) \rho_*(X). \quad (1.9)$$

¹The general linear group $\mathrm{GL}_n(\mathbb{F})$ is the set of $n \times n$ invertible matrices over the field $\mathbb{F}$, together with the operation of ordinary matrix multiplication.

²The result on the right is often written as $[\rho_*(X), \rho_*(Y)]$ using the ordinary commutator. One has hence to be careful to distinguish it from the Lie bracket.

A representation is therefore a concrete realisation of abstract generators as matrices or linear operators. It is **faithful** if different algebra elements give different operators. Much of Lie theory in accelerator physics consists of choosing the representation best suited to the calculation: matrices for linear optics, differential operators for nonlinear maps, or finite truncations acting on Taylor coefficients (TPSAs) for numerical implementations of nonlinear maps.

1.1.3 Example: real 2×2 matrices

Consider

$$G = \mathrm{GL}_2(\mathbb{R}) = \{A \in \mathbb{R}^{2 \times 2} : \det A \neq 0\}. \quad (1.10)$$

This is a Lie group under matrix multiplication. It is not a vector space: the zero matrix is excluded, and two invertible matrices can have a singular sum. A curve through the identity can be written $A(t) = I + tX + \mathcal{O}(t^2)$, where X is an arbitrary real 2×2 (potentially singular) matrix. Hence

$$\mathfrak{gl}(2, \mathbb{R}) = \mathbb{R}^{2 \times 2}, \quad [X, Y] = XY - YX. \quad (1.11)$$

The commutator obeys Eqs. (1.2)–(1.4). Using the four matrices E_{ij} , with a one in row i , column j , and zeros elsewhere, as generators gives

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}. \quad (1.12)$$

For example, $[E_{12}, E_{21}] = E_{11} - E_{22}$. This is the elementary model for the same commutator structure encountered with larger transfer matrices.

1.1.4 Example: $\mathfrak{su}(N)$

The special unitary group is

$$\mathrm{SU}(N) = \{U \in \mathbb{C}^{N \times N} : U^\dagger U = I, \det U = 1\}. \quad (1.13)$$

Put $U(t) = I + tX + \mathcal{O}(t^2)$ and differentiate both defining conditions at $t = 0$. One obtains

$$\mathfrak{su}(N) = \{X \in \mathbb{C}^{N \times N} : X^\dagger = -X, \mathrm{tr} X = 0\}, \quad [X, Y] = XY - YX. \quad (1.14)$$

It is a real vector space of dimension $N^2 - 1$. Physicists often prefer Hermitian generators t_a and write $X_a = -it_a$. With the conventional normalisation,

$$[t_a, t_b] = if_{ab}^{c} t_c, \quad [X_a, X_b] = f_{ab}^{c} X_c. \quad (1.15)$$

For $\mathrm{SU}(2)$ one may take $t_a = \sigma_a/2$, where the σ_a are the Pauli matrices, and $f_{ab}^{c} = \epsilon_{abc}$. This example is useful because it separates the abstract real Lie algebra from two common matrix conventions: anti-Hermitian generators are closed under the ordinary commutator, whereas Hermitian generators introduce an explicit factor of i .

1.2 Hamiltonian Maps in Six-Dimensional Phase Space

The application of Lie algebra to nonlinear accelerator physics takes us one step beyond ordinary matrix spaces, and requires a bit of differential geometry to lay the necessary mathematical foundation. One can skip some of these parts and jump directly to the relevant part on the Lie algebra of Hamiltonian maps in Sec. 1.2.5.

1.2.1 Manifolds

A smooth n -dimensional **manifold** M is a space that looks locally like $\mathbb{R}^n$: every point has a neighbourhood that can be described by n coordinates, and changes between overlapping coordinate systems are smooth. Coordinates therefore let us perform calculations locally, but the coordinate tuple assigned to a point should not be confused with the point itself.

For a particle in three degrees of freedom, the familiar canonical phase space is

$$M = \mathbb{R}^6, \quad z = (q_1, p_1, q_2, p_2, q_3, p_3)^T. \quad (1.16)$$

This particular manifold **is** a vector space: componentwise addition and scalar multiplication remain in $\mathbb{R}^6$, and all vector-space axioms follow component by component from those of $\mathbb{R}$. A general manifold, however, need not be a vector space. For instance, the configuration space of a plane pendulum is the circle S^1 , and its phase space is

$$T^*S^1 \simeq S^1 \times \mathbb{R}, \quad (1.17)$$

which is a cylinder: the angle is periodic, whereas its conjugate momentum is real. By itself, the manifold structure supplies neither addition nor scalar multiplication of points. We could define addition as, for instance,

$$(\theta_1, p_1) + (\theta_2, p_2) = (\theta_1 + \theta_2 \pmod{2\pi}, p_1 + p_2), \quad (1.18)$$

which works beautifully. But notice something subtle: here we have implicitly chosen which point of the circle corresponds to $\theta = 0$ to deal with the periodicity of the angle. And this is the key point: the manifold structure by itself supplies no preferred addition law for points. A coordinate-independent addition law can certainly exist, as it does when the manifold is also a Lie group, but it is **additional structure** rather than part of the definition of a manifold. More fundamentally, every finite-dimensional real vector space is topologically equivalent to some $\mathbb{R}^n$ and can be continuously contracted to a point, whereas a loop around the cylinder cannot be contracted. Thus, being locally Euclidean does not make a manifold a vector space.

1.2.2 Tangent space

Although points of a general manifold cannot be added, the infinitesimal directions at any fixed point can. These directions form the **tangent space** $T_p M$ at $p \in M$. One way to define a tangent vector at p is by a smooth curve $\gamma(t)$ through that point, with $\gamma(0) = p$: two curves represent the same tangent vector if their derivatives at

$t = 0$ agree in one (and hence every) local coordinate system. In local coordinates $(x^1, \dots, x^n)$, a tangent vector can be written

$$v = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i} \Big|_p. \quad (1.19)$$

Its components depend on the chosen coordinates, but the vector itself does not. Crucially, $T_p M$ is an n -dimensional vector space, even when M is not: tangent vectors based at the same point may be added and multiplied by scalars. The collection of tangent spaces at all points is the **tangent bundle**

$$TM = \bigsqcup_{p \in M} T_p M, \quad (1.20)$$

$$= \{(p, v) \mid p \in M, v \in T_p M\}. \quad (1.21)$$

The **cotangent space** $T_p^* M$ is the dual vector space of $T_p M$; its elements are linear maps $T_p M \rightarrow \mathbb{R}$. In the coordinates above it has the dual basis $(dx^1, \dots, dx^n)$. Collecting these dual spaces gives the **cotangent bundle**

$$T^* M = \bigsqcup_{p \in M} T_p^* M, \quad (1.22)$$

$$= \{(p, v^*) \mid p \in M, v^* \in T_p^* M\}. \quad (1.23)$$

This explains the notation used for the pendulum: a point of $T^* S^1$ is a pair (θ, p_θ) , where $\theta \in S^1$ specifies the configuration and $p_\theta d\theta \in T_\theta^* S^1$ is its conjugate momentum.

Practical interpretation in phase space. For accelerator calculations, these abstract spaces have a very concrete meaning. If $z = (q_1, p_1, \dots, q_n, p_n) \in M$ is a phase-space point, then a tangent vector $v \in T_z M$ is represented in canonical coordinates by an infinitesimal phase-space variation

$$\delta z = (\delta q_1, \delta p_1, \dots, \delta q_n, \delta p_n)^T. \quad (1.24)$$

Equivalently, it may be the velocity $\dot{z}(0)$ of a curve through z . Thus the tangent vectors of phase space are **not** the canonical momenta: the p_i are already coordinates of the point $z \in M$, whereas a tangent vector describes how the complete phase-space point changes. For example, the Jacobian of a tracking map acts precisely on such variations, $\delta z_{\text{out}} = D\mathcal{M}_z \delta z_{\text{in}}$.

A cotangent vector $\alpha \in T_z^* M$ is instead a linear rule that assigns a number to each infinitesimal variation. In coordinates,

$$\alpha = \alpha_i dz^i, \quad \alpha(v) = \alpha_i v^i. \quad (1.25)$$

The differential of a scalar function is the most common example: $df = (\partial f / \partial z^i) dz^i$ and $df(v) = v^i \partial_i f$, the directional derivative of f along v . Writing a covector as a row vector or “gradient” is therefore a coordinate representation; identifying it with a vector by a dot product additionally uses a metric and should not be confused with the intrinsic vector–covector duality.

There is one related use of the word momentum that is important but lives one level lower. If the phase space itself is a cotangent bundle $M = T^*Q$, then a phase-space point is (q, p) with $q \in Q$ and $p \in T_q^*Q$. In that sense the canonical momenta are the components of a covector on the **configuration space** Q ; they are still not tangent vectors of the phase space M .

1.2.3 Symplectic manifolds

Phase space carries yet another layer of structure. At a point $p \in M$, an alternating covariant 2-tensor is an element

$$\omega_p \in \bigwedge^2 T_p^*M. \quad (1.26)$$

Here $\bigwedge^2 T_p^*M$ is the vector space of antisymmetric bilinear functions

$$\omega_p : T_pM \times T_pM \longrightarrow \mathbb{R}, \quad \omega_p(u, v) = -\omega_p(v, u). \quad (1.27)$$

A **differential 2-form** ω on M is a choice of such a tensor ω_p at every p that varies smoothly with p . The vector space of all differential 2-forms on M is denoted by $\Omega^2(M)$. The pair (M, ω) is a **symplectic manifold** if ω is closed and non-degenerate:

$$d\omega = 0, \quad (1.28)$$

$$\omega_p(v, w) = 0 \quad \forall w \in T_pM \iff v = 0. \quad (1.29)$$

Here $d\omega$ is the exterior derivative of ω , and the second condition is required at every $p \in M$. An antisymmetric bilinear form can be non-degenerate only in even dimensions. Darboux's theorem guarantees that, near every point of a $2n$ -dimensional symplectic manifold, there are canonical coordinates in which

$$\omega = \sum_{j=1}^n dq_j \wedge dp_j. \quad (1.30)$$

For the pendulum this is $\omega = d\theta \wedge dp_\theta$; the same local formula works even though θ is not a global real coordinate. For the special case $M = \mathbb{R}^6$, there is a natural identification

$$T_z\mathbb{R}^6 \simeq \mathbb{R}^6. \quad (1.31)$$

at every z . Therefore, we can write

$$\omega : \mathbb{R}^6 \times \mathbb{R}^6 \rightarrow \mathbb{R}, \quad \omega(u, v) = u^T S v, \quad (1.32)$$

with

$$S = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}. \quad (1.33)$$

We can hence think of S as the coordinate representation of ω .

1.2.4 Symplectic maps

A canonical, or symplectic, map $\mathcal{M} : M \rightarrow M$ is a diffeomorphism that preserves the symplectic form. The easiest way to understand this is to follow two infinitesimal phase-space variations. At $z \in M$, the differential (or **pushforward**)

$$D\mathcal{M}_z : T_z M \longrightarrow T_{\mathcal{M}(z)} M. \quad (1.34)$$

propagates a tangent vector u at the entrance into the tangent vector $D\mathcal{M}_z u$ at the image point. In accelerator language this is the linearised tracking map about the reference point z .

The symplectic form at the image point can therefore be evaluated on the two transported vectors,

$$\omega_{\mathcal{M}(z)}(D\mathcal{M}_z u, D\mathcal{M}_z v). \quad (1.35)$$

This quantity depends on the original point z and on the original vectors $u, v \in T_z M$, so it defines a new 2-form at z . That new form is, by definition, the **pullback** of ω by $\mathcal{M}$:

$$(\mathcal{M}^* \omega)_z(u, v) \equiv \omega_{\mathcal{M}(z)}(D\mathcal{M}_z u, D\mathcal{M}_z v). \quad (1.36)$$

Thus $D\mathcal{M}$ pushes vectors forward, while $\mathcal{M}^*$ pulls forms back so that they can again be compared at the original point. The map is symplectic precisely when this comparison shows no change,

$$\mathcal{M}^* \omega = \omega, \quad (1.37)$$

or, point by point,

$$\omega_{\mathcal{M}(z)}(D\mathcal{M}_z u, D\mathcal{M}_z v) = \omega_z(u, v), \quad (1.38)$$

for every z and every $u, v \in T_z M$. Nothing physically moves “backwards” in a pull-back: it is simply the bookkeeping operation that expresses a form from the output point in terms of input tangent vectors.

In canonical coordinates on the $2n$ -dimensional phase space, let $J(z) : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be the Jacobian matrix representing $D\mathcal{M}_z$, with entries $J_{ij}(z) = \partial \mathcal{M}_i / \partial z_j$. Thus, if $\mathbf{u}$ is the coordinate vector representing $u \in T_z M$, then $J(z)\mathbf{u}$ represents $D\mathcal{M}_z u \in T_{\mathcal{M}(z)} M$. Since the matrix of ω is the constant matrix S in these coordinates, the preceding identity becomes, for arbitrary coordinate vectors $\mathbf{u}$ and $\mathbf{v}$,

$$(J(z)\mathbf{u})^T S (J(z)\mathbf{v}) = \mathbf{u}^T S \mathbf{v}. \quad (1.39)$$

Equivalently,

$$\mathbf{u}^T (J(z)^T S J(z) - S) \mathbf{v} = 0, \quad (1.40)$$

for all $\mathbf{u}$ and $\mathbf{v}$. The matrix in parentheses must therefore vanish, which gives the familiar Jacobian condition

$$J(z)^T S J(z) = S. \quad (1.41)$$

Consequently, in canonical coordinates, Eq. (1.41) is equivalent to the geometric symplecticity condition (1.37).

For a **linear** map $z \mapsto Rz$, the Jacobian is simply the constant matrix R . The linear symplectic maps therefore form the **symplectic group**

$$\text{Sp}(2n, \mathbb{R}) = \{R \in \text{GL}(2n, \mathbb{R}) \mid R^T S R = S\}. \quad (1.42)$$

This is the group containing the familiar linear transfer matrices of Hamiltonian beam optics. To find its Lie algebra, write a curve through the identity as $R(t) = I + tA + \mathcal{O}(t^2)$ and differentiate $R(t)^T S R(t) = S$ at $t = 0$. One obtains

$$\mathfrak{sp}(2n, \mathbb{R}) = \{A \in \mathbb{R}^{2n \times 2n} \mid A^T S + S A = 0\}. \quad (1.43)$$

Equivalently, SA is symmetric. This provides the direct linear analogue of the nonlinear construction below: a symplectic matrix is a finite canonical transformation, while an element of $\mathfrak{sp}(2n, \mathbb{R})$ is its infinitesimal generator. For quadratic Hamiltonians the Hamiltonian vector field is linear, so exponentiating that vector field reproduces precisely a matrix in $\mathrm{Sp}(2n, \mathbb{R})$. Higher-order Hamiltonians are the nonlinear extension of the same idea.

1.2.5 Hamiltonian Lie algebra and Poisson algebra

The infinitesimal generators of Hamiltonian maps are smooth functions on phase space. Let $C^\infty(M)$ denote all smooth real-valued functions on M . It is a vector space under pointwise operations,

$$(af + bg)(z) = af(z) + bg(z). \quad (1.44)$$

Indeed, a linear combination of smooth functions is smooth, so the set is closed; all the remaining vector-space laws follow from the corresponding laws for real numbers at every point z . This is also an infinite-dimensional vector space. For example, on a canonical coordinate patch the functions $1, q_1, q_1^2, \dots$ (multiplied by a smooth cutoff if the patch is not global) contain arbitrarily large linearly independent subsets.

The relevant functions here are **smooth Hamiltonians**, not necessarily “integrable functions.” An integrable Hamiltonian in the Liouville sense is a Hamiltonian system with sufficiently many independent invariants in involution; such Hamiltonians do not form the vector space used below. Likewise, mere integrability in the measure-theoretic sense would not provide the derivatives needed for a Poisson bracket.

In canonical coordinates the Poisson bracket is

$$\llbracket f, g \rrbracket_P = \sum_{j=1}^n \left(\frac{\partial f}{\partial q_j} \frac{\partial g}{\partial p_j} - \frac{\partial f}{\partial p_j} \frac{\partial g}{\partial q_j} \right). \quad (1.45)$$

This is the convention used by Dragt and Forest: the first argument is differentiated with respect to the coordinate q_j in the first term. In particular,

$$\llbracket q_i, p_j \rrbracket = \delta_{ij}, \quad \llbracket q_i, q_j \rrbracket = \llbracket p_i, p_j \rrbracket = 0. \quad (1.46)$$

It is bilinear and antisymmetric, satisfies the Jacobi identity, and maps two smooth functions to another smooth function. Therefore $(C^\infty(M), \llbracket \cdot, \cdot \rrbracket_P)$ is a Lie algebra. A constant has zero Poisson bracket with every function and generates no motion, so Hamiltonians that differ only by a constant describe the same infinitesimal map. On a connected phase space the non-redundant algebra is consequently

$$\mathfrak{ham}(M) \simeq C^\infty(M)/\mathbb{R}. \quad (1.47)$$

This quotient is again a vector space: define $[f] + [g] = [f + g]$ and $a[f] = [af]$, where $[f]$ denotes all functions that differ from f by a constant. Changing representatives only changes the result by a constant, so these operations are well-defined. The same argument and $[[f + c, g + d]] = [[f, g]]$ show that the Poisson bracket is well-defined on the quotient. The algebra remains infinite-dimensional. With the conventions used here, the bracket-preserving identification with the tangent vector fields is

$$[f] \mapsto -X_f, \quad [-X_f, -X_g] = -X_{[[f, g]]}. \quad (1.48)$$

One may instead identify $[f]$ with X_f . The corresponding function-space bracket then acquires an overall minus sign. This harmless choice is the origin of several sign conventions in the Lie-map literature.

There is a small global qualification behind Eq. (1.47). The tangent algebra to all symplectic maps consists of symplectic vector fields; a symplectic vector field need only be locally Hamiltonian. On the usual $\mathbb{R}^6$ phase space every such field of interest is generated by a global Hamiltonian, while on a manifold with non-trivial topology the Hamiltonian maps can form a proper subgroup of all symplectic maps. For local accelerator maps, $C^\infty(M)/\mathbb{R}$ is the algebra normally used.

The full function space has still more structure. With ordinary pointwise multiplication it is a commutative associative algebra, and the bracket satisfies the Leibniz rule

$$[[f, gh]] = [[f, g]]h + g[[f, h]]. \quad (1.49)$$

A commutative algebra equipped with a Lie bracket satisfying Eq. (1.49) is called a **Poisson algebra**. Thus $C^\infty(M)$ is both a commutative algebra and a Lie algebra, with the two products tied together by the Leibniz rule. Strictly speaking, the quotient $C^\infty(M)/\mathbb{R}$ is a Lie algebra but not a Poisson algebra under pointwise multiplication, because constants are not an ideal for that multiplication. In calculations one therefore keeps representatives in $C^\infty(M)$ and discards irrelevant constant terms when interpreting the generator.

This Poisson-algebra structure is particularly useful for accelerator physics. It permits brackets of polynomial Hamiltonians to be evaluated by ordinary differentiation and multiplication, it extends immediately to Taylor and truncated-power-series representations, and the Leibniz rule controls the action of a generator on products of observables. At the same time, working with a Hamiltonian generator rather than arbitrary Taylor coefficients keeps the underlying map canonical.

1.3 Exponentiation

1.3.1 From an infinitesimal generator to a finite transformation

For a Lie group G with algebra $\mathfrak{g}$, the exponential map

$$\exp : \mathfrak{g} \longrightarrow G. \quad (1.50)$$

is defined through one-parameter subgroups. For every $X \in \mathfrak{g}$ there is a unique one-parameter subgroup $\gamma_X : \mathbb{R} \rightarrow G$ with

$$\gamma_X(0) = e, \quad \dot{\gamma}_X(0) = X, \quad \gamma_X(t+s) = \gamma_X(t)\gamma_X(s). \quad (1.51)$$

The parameter is $t \in \mathbb{R}$, not merely $t \geq 0$: negative values give the inverse transformation, $\gamma_X(-t) = \gamma_X(t)^{-1}$. At this abstract stage t is only a group parameter; depending on the application it may later be physical time, longitudinal position, an angle, a magnet strength multiplied by a length, or simply a dimensionless interpolation parameter. One defines

$$\exp(tX) \equiv \gamma_X(t), \quad \exp(X) = \gamma_X(1). \quad (1.52)$$

Thus exponentiation means following the same infinitesimal generator X for a finite amount of its parameter. This geometric construction is the definition; a power series appears only after choosing a representation in which the generator is a linear operator.

For a matrix Lie group, X is a matrix and the group exponential is exactly the convergent matrix series

$$\exp X = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots. \quad (1.53)$$

More generally, let

$$\rho : G \rightarrow \mathrm{GL}(V). \quad (1.54)$$

be a representation of the group. Its differential at the identity,

$$\rho_* \equiv (d\rho)_e : \mathfrak{g} \rightarrow \mathfrak{gl}(V), \quad (1.55)$$

is the corresponding Lie-algebra representation introduced above. It maps the abstract infinitesimal generator X to the linear operator $\rho_*(X)$ acting on V . Group exponentiation and operator exponentiation then commute with the representation:

$$\rho(\exp tX) = \exp(t\rho_*(X)). \quad (1.56)$$

This equation is not an additional definition: it states that representing a finite group transformation gives the same result as first representing its generator and then exponentiating that operator.

For bounded operators the analogous operator series converges. For unbounded differential operators, or for formal Taylor maps, its domain and convergence require care and the series is often used order by order as a formal series. Moreover, the exponential need not be one-to-one or cover the entire group, although for a finite-dimensional Lie group its image contains a neighbourhood of the identity.

1.3.2 Example: $\mathrm{SU}(N)$

If $X \in \mathfrak{su}(N)$, Eq. (1.53) gives an element of $\mathrm{SU}(N)$. Indeed,

$$(e^X)^\dagger e^X = e^{X^\dagger} e^X = e^{-X} e^X = I, \quad \det(e^X) = e^{\mathrm{tr} X} = 1. \quad (1.57)$$

With Hermitian physics generators this is commonly written

$$U(\theta) = \exp(-i\theta^a T_a). \quad (1.58)$$

The generator is linear in the parameters, while the exponential builds the finite, generally non-commuting transformation.

1.3.3 Example: Hamiltonian flow

The preceding abstract construction becomes especially concrete in phase space. Let $f \in C^\infty(M)$. Its Hamiltonian vector field X_f is the vector field defined, for the symplectic convention used here, by

$$\iota_{X_f}\omega = df, \quad (1.59)$$

where ι_{X_f} denotes insertion of X_f into the first argument of the 2-form. In canonical coordinates this gives

$$X_f = \sum_{j=1}^n \left(\frac{\partial f}{\partial p_j} \frac{\partial}{\partial q_j} - \frac{\partial f}{\partial q_j} \frac{\partial}{\partial p_j} \right), \quad X_f g = \llbracket g, f \rrbracket = -\llbracket f, g \rrbracket. \quad (1.60)$$

Thus $X_f(z) \in T_z M$ gives an infinitesimal direction of motion at every phase-space point.

The **flow** of this vector field is the family of point maps

$$\Phi_f^t : M \rightarrow M. \quad (1.61)$$

defined by the initial-value problem

$$\frac{d}{dt}\Phi_f^t(z) = X_f(\Phi_f^t(z)), \quad \Phi_f^0(z) = z. \quad (1.62)$$

For each initial point z , the curve $t \mapsto \Phi_f^t(z)$ is the integral curve of X_f through z . If $f = H$ is the physical Hamiltonian and t is physical time, Eq. (1.62) is simply Hamilton's equations,

$$\dot{q}_j = \frac{\partial H}{\partial p_j}, \quad \dot{p}_j = -\frac{\partial H}{\partial q_j}, \quad (1.63)$$

or, for any observable g with no explicit t dependence,

$$\frac{dg}{dt} = X_H g = \llbracket g, H \rrbracket. \quad (1.64)$$

In accelerator applications the evolution parameter need not be time; after a suitable Hamiltonian formulation it is often the longitudinal coordinate s . Mathematically nothing changes.

For an autonomous generator, the flow satisfies

$$\Phi_f^{t+s} = \Phi_f^t \circ \Phi_f^s, \quad (\Phi_f^t)^{-1} = \Phi_f^{-t}, \quad (1.65)$$

whenever the relevant solutions exist. It is therefore a one-parameter (local, or global when the vector field is complete) group of diffeomorphisms, and the group exponential is its unit-parameter map,

$$\exp(tX_f) = \Phi_f^t, \quad \exp(X_f) = \Phi_f^1. \quad (1.66)$$

This is the precise meaning of exponentiating a Hamiltonian vector field.

The flow is symplectic. Cartan's formula and Eq. (1.59) give

$$\mathcal{L}_{X_f}\omega = d(\iota_{X_f}\omega) + \iota_{X_f}(d\omega) = d^2f = 0. \quad (1.67)$$

Using the standard identity for differentiation of a pulled-back form along a flow,

$$\frac{d}{dt}(\Phi_f^t)^*\omega = (\Phi_f^t)^*(\mathcal{L}_{X_f}\omega) = 0, \quad (1.68)$$

so $(\Phi_f^t)^*\omega = \omega$ because Φ_f^0 is the identity. In words: infinitesimal Hamiltonian motion preserves the symplectic form, hence so does the finite map obtained by following that motion.

If the generator depends explicitly on the evolution parameter, $f = f_t$, the maps from different parameter values are generated by different vector fields and do not in general form $\exp(tX_f)$ for one fixed X_f . The solution is then a parameter-ordered exponential (or, equivalently, the solution of the non-autonomous flow equation). This is analogous to a sequence of non-commuting beam-line elements, which cannot generally be replaced by the exponential of their simple sum.

1.3.4 The adjoint representation

Every Lie group has a natural representation on its own algebra. Conjugate a group element close to the identity by $g \in G$ and differentiate at the identity:

$$\text{Ad}_g X = \left. \frac{d}{dt} \right|_{t=0} g \exp(tX) g^{-1}. \quad (1.69)$$

This is the **adjoint representation** of the group. For a matrix group,

$$\text{Ad}_g X = gXg^{-1}. \quad (1.70)$$

Differentiating once more gives the adjoint representation of the algebra,

$$\text{ad}_X : \mathfrak{g} \longrightarrow \mathfrak{g}, \quad \text{ad}_X(Y) = [X, Y]. \quad (1.71)$$

The Jacobi identity is exactly the statement

$$[\text{ad}_X, \text{ad}_Y] = \text{ad}_{[X, Y]}. \quad (1.72)$$

Thus the adjoint operators form a Lie algebra under the operator commutator. More precisely, ad_X is an element of $\mathfrak{gl}(\mathfrak{g})$, not in general an element of the original $\mathfrak{g}$; the image $\text{ad}(\mathfrak{g})$ is a Lie subalgebra of operators. Elements in the centre of $\mathfrak{g}$ act as zero, so the adjoint representation is faithful precisely when the centre is trivial.

Exponentiating this representation gives

$$\text{Ad}_{\exp X} = \exp(\text{ad}_X) = I + \text{ad}_X + \frac{\text{ad}_X^2}{2!} + \cdots. \quad (1.73)$$

This is now an ordinary exponential series of linear operators acting on algebra elements. For $\text{SU}(N)$,

$$\text{Ad}_U(Y) = UYU^\dagger, \quad \text{ad}_X(Y) = [X, Y], \quad (1.74)$$

and Eq. (1.73) states $e^X Y e^{-X} = e^{\text{ad}_X} Y$. In the generator basis $X_a = -iT_a$, the matrices of the adjoint representation have components $(\text{ad}_{X_a})_b^c = f_{ab}^c$.

1.3.5 Lie operators on phase-space functions

The adjoint representation of the Hamiltonian algebra is the Lie operator used in non-linear beam dynamics:

$$:f:g \equiv \text{ad}_f(g) = \llbracket f, g \rrbracket. \quad (1.75)$$

It is a linear differential operator on functions, and the Jacobi identity gives

$$[:f:, :g:] = :\llbracket f, g \rrbracket:. \quad (1.76)$$

Constants disappear automatically, so this is a representation of $C^\infty(M)/\mathbb{R}$. The exponentiated Lie operator is

$$e^{:f:}g = g + \llbracket f, g \rrbracket + \frac{1}{2!}\llbracket f, \llbracket f, g \rrbracket \rrbracket + \cdots. \quad (1.77)$$

Unlike the abstract definition of the group exponential, this expression is an exponential power series: it is the operator exponential in the adjoint representation.

The relation to the point flow is now immediate. With the Dragt–Forest convention of Eq. (1.45),

$$:f:g = \llbracket f, g \rrbracket = -X_f g. \quad (1.78)$$

Therefore the operator exponential acts on a function by pullback along the **inverse** Hamiltonian flow,

$$e^{:f:}g = g \circ \Phi_f^{-t} = (\Phi_f^{-t})^* g. \quad (1.79)$$

In particular, applying it componentwise to the coordinate functions gives

$$\Phi_H^t(z) = e^{-t:H:}z. \quad (1.80)$$

This minus sign is not a dynamical convention hidden in Hamilton's equations; it comes solely from defining the Lie operator with the generator in the **first** Poisson-bracket slot, $:f:g = \llbracket f, g \rrbracket$, whereas the Hamiltonian vector field acts as $X_f g = \llbracket g, f \rrbracket$. This is the Dragt–Forest convention used throughout this chapter.

Because of the Leibniz rule, $:f:$ is a derivation. Its exponential is therefore an algebra automorphism:

$$e^{:f:}(gh) = (e^{:f:}g)(e^{:f:}h), \quad \llbracket e^{:f:}g, e^{:f:}h \rrbracket = e^{:f:}\llbracket g, h \rrbracket. \quad (1.81)$$

Acting on the coordinate functions gives a canonical map,

$$\mathcal{M}_f(z) = e^{:f:}z = \Phi_f^{-1}(z), \quad (1.82)$$

where the exponential acts on every component of z and the final equality refers to unit flow parameter. Thus a Lie map with generator f is the inverse unit-parameter flow of the Hamiltonian vector field X_f under the convention adopted here. Products of these exponentials are the **compositional maps** used in Lie factorisations [1, 2]. Non-commutativity is retained through the Baker–Campbell–Hausdorff series; formally,

$$e^{:f:}e^{:g:} = e^{:h:}, \quad h = f + g + \frac{1}{2}\llbracket f, g \rrbracket + \cdots, \quad (1.83)$$

up to an irrelevant additive constant in h .

1.3.6 Why map order can appear to be reversed

Two related objects are frequently denoted by the same symbol in the accelerator literature:

1. a **point map** $\mathcal{A} : M \rightarrow M$, which takes a phase-space point as its argument; and
2. the corresponding **coordinate-substitution operator** $\hat{\mathcal{A}}$, which acts on a function according to

$$\hat{\mathcal{A}}g = g \circ \mathcal{A}. \quad (1.84)$$

Even if $\mathcal{A}$ is nonlinear, $\hat{\mathcal{A}}$ is a linear operator on the vector space of functions. In particular, $\widehat{\mathcal{M}_f} = e^{\mathcal{M}_f}$.

Suppose a particle first passes through $\mathcal{A}$ and then through $\mathcal{B}$. For point maps with explicit arguments,

$$z_1 = \mathcal{A}(z_0), \quad z_2 = \mathcal{B}(z_1) = (\mathcal{B} \circ \mathcal{A})(z_0). \quad (1.85)$$

The rightmost point map acts first. Matrices multiplying a column vector follow the same rule: $z_2 = B A z_0$.

Substitution operators compose in the opposite order:

$$\widehat{\mathcal{B} \circ \mathcal{A}}g = g \circ \mathcal{B} \circ \mathcal{A} = \hat{\mathcal{A}}\hat{\mathcal{B}}g. \quad (1.86)$$

For the coordinate functions themselves this becomes

$$z_2 = (\hat{\mathcal{A}}\hat{\mathcal{B}}z)(z_0). \quad (1.87)$$

The labels in the operator product therefore follow the physical beam-line order from left to right, even though, as operators acting on the object to their right, the rightmost operator is evaluated first. This pullback reversal is the source of the apparently conflicting conventions.

In the notation often associated with Dragt, products of Lie transformations such as

$$e^{\mathcal{M}_A} e^{\mathcal{M}_B} z. \quad (1.88)$$

are coordinate-substitution operators, so A is encountered before B in the corresponding physical sequence. If instead one writes maps with their phase-space arguments, the same sequence is

$$\mathcal{M}_B(\mathcal{M}_A(z)) = (\mathcal{M}_B \circ \mathcal{M}_A)(z). \quad (1.89)$$

There is no disagreement about composition: one expression acts on functions (including the coordinate functions), while the other acts on points. Whenever an ordering is ambiguous, displaying either the argument or the composition symbol $\circ$ removes it.

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